PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Inventra

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1.0 Project Overview

1.1 Problem Statement

Malaysia has about 1.1 million MSMEs, contributing around 40% of the country's GDP, with many operating in retail and food and beverage, including about 476,000 retail businesses and 136,000 F&B businesses in 2022. This shows that small retail and food businesses are important to Malaysia's economy, yet many grocery kiosks and small retail chains still rely on basic POS reports, manual stock checks, spreadsheets, and personal judgment to manage inventory and outlet decisions. As a result, they often face stock shortages, excess stock, wasted money, expired items, and weak expansion planning. This is especially serious for food-related businesses, as Malaysia produces about 38,000 tonnes of solid waste per day, including about 17,000 tonnes of food waste. Therefore, there is a clear need for an AI-powered decision support system that helps small businesses manage inventory better, predict demand, and make smarter expansion decisions.

1.2 Target Domain

This solution focuses on helping grocery kiosks and small retail chains make better business decisions through AI-powered inventory, demand, and expansion intelligence. It supports SME operators in improving stock planning, reducing waste, managing working capital more efficiently, and evaluating new outlet opportunities using data instead of guesswork. This fits the hackathon's Domain 2 focus on building AI systems that generate insights, recommend actions, and improve economic outcomes through smarter decision-making.

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1.3 Proposed Solution Summary

Our proposed solution is an AI-powered decision intelligence platform for grocery kiosks and small retail chains that combines inventory visibility, demand forecasting, and location planning in one system. It provides a 3D inventory view for monitoring stock across shelves and locations, a forecasting dashboard for restocking recommendations, an AI chatbot with RAG for inventory-related assistance, and a multi-agent location strategy module for comparing potential outlets. Overall, the platform helps SMEs move from manual and reactive decision-making to a more data-driven and forward-looking approach in managing daily operations and business growth.

2.0 Background & Business Objective

2.1 Background of the Problem

Many small businesses already use POS systems or basic digital tools, but these tools mainly help them record sales and stock, not make better decisions. Inventory data, sales history, and outlet performance are often separated, so owners cannot easily see what products need to be restocked, what items are moving slowly, or which outlet is performing better.

This becomes a bigger problem in retail and food businesses because demand can change by location, season, and time. Without proper forecasting and inventory planning, businesses may order too little and lose sales, or order too much and waste money on stock that may expire or stay unsold.

The same issue happens when businesses want to expand. Many owners still choose new locations based on guesswork instead of data such as customer demand, competition, rent, and financial risk. This makes expansion more risky, especially for small businesses with limited capital.

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2.2 Importance of Solving This Issue

Solving this issue is important because MSMEs are a major part of Malaysia's economy. If small businesses can make better stock and expansion decisions, they can reduce waste, improve cash flow, avoid missed sales, and grow more safely. Since MSMEs contribute around 40% of GDP, improving their efficiency can also bring wider economic benefits.

This is also important for food-related businesses because poor stock planning can lead to unnecessary food waste. In a country where about 17,000 tonnes of food waste are produced every day, helping businesses manage stock more accurately can reduce both financial losses and wasted resources.

Most importantly, solving this problem helps businesses move from making decisions based on instinct to making decisions based on data. That gives owners more confidence in daily operations and future growth.

2.3 Strategic Fit / Impact

This solution fits the hackathon domain well because it is not just an automation tool. It is a decision intelligence system that uses AI to study business data, forecast demand, suggest restocking actions, and support location planning. This matches the Domain 2 requirement, which focuses on helping users make smarter and more informed economic decisions with AI.

The expected impact is clear. The system can help reduce stock shortages, reduce overstocking, lower expiry waste, improve use of working capital, and support better outlet expansion decisions. For users, this means less time spent checking stock manually, less guesswork, and more confidence when planning what to buy, where to expand, and how to manage each location.

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3.0 Product Purpose

3.1. Main Goal of the System

- To provide grocery kiosk operators and small retail businesses with an AI-powered decision support platform for inventory, replenishment, and location planning.
- To improve business decision-making by transforming sales and inventory data into forecasting insights and actionable recommendations.
- To enable more efficient stock management, better replenishment planning, and more informed outlet expansion decisions.
- To reduce operational inefficiencies such as stock shortages, excess inventory, expiry waste, and poor location choices.
- To support SMEs in adopting a more data-driven, efficient, and explainable approach to operations and growth.

3.2. Intended Users (Target Audience)

User Group	Primary Need / Expected Value
Kiosk / Store Owners	Need a fast summary of what to restock, what to reduce, and which outlet needs attention.
Inventory / Operations Managers	Need outlet-level stock visibility, low-stock alerts, expiry monitoring, and recommended reorder quantities.
Procurement Staff	Need an exportable shopping list and a clear justification for replenishment quantities.

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Outlet Supervisors	Need store-specific inventory tracking, stock health by shelf, and location switching for daily operations.
Expansion / Finance Leads	Need side-by-side location comparison, projected demand, cash requirement estimates, and risk scoring.

4. System Functionalities

4.1. Description

Inventra operates as a decision intelligence platform layered above POS and inventory data. It ingests sales and stock signals from connected sources, maps them to kiosk layouts, forecasts short-term demand, retrieves relevant history for natural-language queries, and orchestrates multi-agent analyses for expansion decisions. Z.AI GLM is the core reasoning engine that converts business context into recommendations, explanations, and action priorities.

4.2. Key Functionalities:

1. **3D Inventory Management & Monitoring**
   The system provides a 3D top-down view of the grocery kiosk environment that enables operators to monitor inventory across shelves, racks, and multiple kiosk locations in a more intuitive manner. Users are able to inspect stock levels, identify stock health through color-coded indicators, search for specific items, review stock history, detect low-stock situations, and monitor expiry-related risks. This functionality improves real-time inventory visibility and supports faster operational decision-making.

2. **Demand Forecasting & Inventory Recommendation Dashboard**
   The system generates short-term demand forecasts for the next seven days based on item-level and location-level data. It presents recommendations on whether stock for

each item should be increased, reduced, or maintained, together with suggested replenishment quantities. The dashboard also displays forecast confidence and seasonality indicators to help users understand upcoming demand patterns. In addition, the system allows forecast results to be exported as a shopping list to support replenishment planning and execution.

3. **AI Inventory Assistant with RAG**

   The system includes an AI-powered inventory assistant that uses retrieval-augmented generation to answer inventory-related business questions in natural language. By retrieving relevant historical POS and inventory data, the assistant can provide demand-related suggestions, recommend restocking priorities, and explain the reasoning behind its outputs in a clear and user-understandable manner. Where relevant, the assistant may also present supporting charts or trend visualizations to strengthen business understanding and confidence.

4. **Location Strategy AI (Multi-Agent)**

   The system includes a multi-agent location strategy module to support data-driven outlet expansion decisions. This functionality evaluates candidate locations by collecting relevant location signals, forecasting potential demand, estimating break-even and cash requirements, assessing operational risk, and generating an overall recommendation. The system presents side-by-side comparisons across multiple potential locations and provides actionable next steps to help business owners make more informed expansion decisions

4.3. AI Model & Prompt Design

4.3.1 Model Selection

Z.AI GLM is selected as the mandatory central model because this product depends on context-aware reasoning across structured business metrics, unstructured operating notes, and comparative decision scenarios. The platform needs more than raw automation: it must explain

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why an item should be restocked, why a location is risky, and why the user should trust the recommendation. GLM is therefore placed at the centre of the insight-generation layer rather than treated as a generic add-on.

4.3.2 Prompting Strategy

The prompting approach is hybrid and task-specific. For the inventory assistant, the system uses retrieval-augmented generation (RAG) so that only relevant historical sales, stock movements, and outlet summaries are provided to GLM before answering. For inventory recommendation summaries, the system uses structured prompts that require a controlled, action-oriented output format. For location strategy, the product uses a multi-step agentic flow in which each agent focuses on one responsibility and the final agent synthesises the recommendation into comparison cards and next-step advice.

4.3.3 Context & Input Handling

- Supported inputs include POS transaction records, manual CSV uploads, outlet inventory states, item catalogues, and candidate location descriptors such as rent level, nearby competitors, and income indicators.
- For the preliminary MVP, manual CSV upload is the guaranteed ingestion path; direct connector coverage for Shopify or Square may be implemented as partial live integration or simulated integration depending technical readiness.
- To control latency and cost, the application caps the assembled prompt context to an internal budget of approximately 8,000 tokens per major reasoning call. Larger datasets are pre-aggregated by location, category, and date before the most relevant chunks are retrieved.
- Oversized or malformed inputs are summarised, chunked, or rejected with a clear error message and data-format guidance rather than silently producing unreliable output.

4.3.4 Fallback & Failure Behavior

- The system displays confidence cues and cites the underlying business context where possible, especially for recommendation and comparison outputs.
- When GLM returns an off-topic or low-confidence answer, the product falls back to dashboard metrics or raw historical summaries and asks the user to narrow the query.

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- When retrieved evidence is insufficient, the system returns an explicit insufficient-data state rather than fabricating a confident answer.
- High-impact actions remain human-in-the-loop. The system recommends but does not automatically purchase stock, place supplier orders, or approve a new outlet opening in the MVP.

5.0 User Stories & Use Cases

5.1 User Stories

Name: Ahmad Rahman

Occupation: Grocery Kiosk Business Owner

Demographic: 35 years old, lives in Kuala Lumpur, Malaysia. Operates a small grocery and food kiosk business with several outlet locations. Has a moderate income level and manages the business together with a small team of employees.

Ahmad's story:
Ahmad owns a growing grocery kiosk business that serves daily essentials and ready-to-eat food items. His outlets are doing well, but as the business grows, it becomes harder for him to manage stock, monitor each kiosk, and decide what products need to be restocked. He also wants to expand to new locations, but most of his decisions are still based on manual checking, POS reports, spreadsheets, and personal judgment. Although he has sales data, he does not have a smart system that can turn that data into clear recommendations for inventory planning and business growth.

Challenges:

- Managing stock across multiple kiosk locations
- Detecting low-stock and overstock situations early
- Reducing waste caused by expired or slow-moving items
- Predicting what items should be prepared for upcoming demand

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- Making expansion decisions without enough data on demand, competition, and financial risk

What Ahmad needs:

Ahmad needs a simple and reliable system that helps him see stock health across all outlets, forecast short-term demand, and receive clear restocking recommendations. He also needs an AI assistant that can answer business questions in an easy way, as well as a location planning tool that helps him compare possible outlets before expanding. Most importantly, he needs one platform that helps him and his employees make faster, smarter, and more confident decisions using business data instead of guesswork.

5.2 Use Cases (Main Interactions)

Use Case	Flow Summary
Inventory Monitoring Flow	User selects a kiosk location, opens the 3D layout, clicks a shelf or rack, reviews stock status and alerts, searches an item, and checks stock history or expiry warnings.
Forecast Review Flow	User opens the forecasting dashboard, filters by location, reviews items marked as Add / Reduce / OK, checks suggested quantities and confidence, and exports a shopping list for replenishment.
AI Assistant Flow	User asks a planning question, the system retrieves relevant inventory and POS history, GLM produces a data-backed explanation, and the interface returns recommended actions with charts when useful.

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Location Evaluation Flow	User enters candidate locations, the agents process location and business signals, the system generates comparison cards, and the user reviews final recommendations and next-step guidance.

Use Case Diagram

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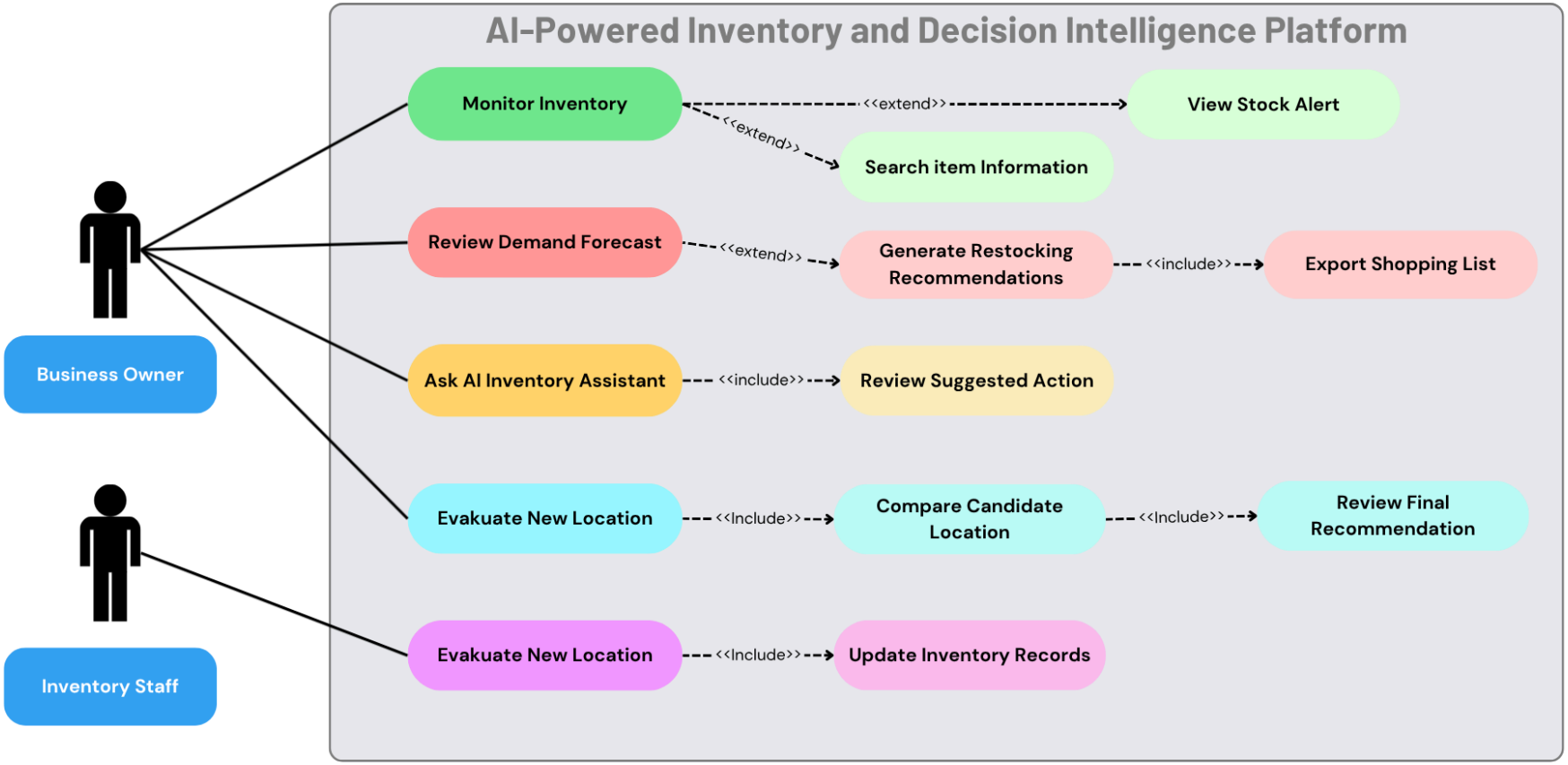

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6.0 Features Included (Scope Definition)

For the UMHackathon preliminary round, the team will prioritise an integrated MVP that clearly demonstrates decision intelligence rather than attempting full production depth. Features are grouped below by implementation priority.

Priority	Feature	Scope in Preliminary MVP
P1 - Must Demo	Multi-location overview dashboard	Overview of outlets, stock status, low-stock count, forecast summary, and quick navigation into decision modules.
P1 - Must Demo	3D inventory map	Simplified top-down store layout with clickable shelves, stock heat indicators, item search, and low-stock / expiry highlighting.
P1 - Must Demo	7-day demand forecasting dashboard	Add / Reduce / OK recommendation board with quantity guidance, seasonality markers, confidence score, and shopping list export.
P1 - Must Demo	AI inventory assistant with RAG	Natural-language query experience backed by historical POS and inventory retrieval for operational planning questions.

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P1 - Must Demo	Location strategy AI	Agent-based comparison of up to five candidate outlets with recommendation cards and action-oriented breakdowns.
P2 - Supporting	POS ingestion and integration management	Manual CSV upload as the required path, plus connector status screens and optional Shopify / Square integration depending technical readiness.
P2 - Supporting	Sync logs and alert centre	View recent integration activity, low-stock warnings, and expiry notifications in one place.

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7.0 Features Not Included (Scope Control)

- Automatic stock purchasing, supplier ordering, or direct payment execution.
- Full ERP, warehouse, or accounting-suite integration beyond the scoped POS and CSV data inputs.
- Real-time IoT sensor, CCTV, or computer-vision shelf monitoring.
- Photorealistic 3D modelling or unlimited custom store-layout editors.
- Native mobile applications for Android or iOS in the preliminary round.
- Advanced role-based access control, enterprise-grade audit logs, or multi-tenant administration.
- Fully autonomous final approval for outlet expansion without human review.
- Guaranteed real-time live sync coverage for every Shopify or Square edge case in the hackathon build.

8.0 Assumptions & Constraints

8.1 LLM Cost Constraint

- GLM calls are concentrated on recommendation reasoning, explanation, chatbot interactions, and location comparison synthesis rather than every dashboard refresh.
- Repeated questions and previously generated summaries are cached to reduce duplicate inference cost.
- Prompt context is restricted through retrieval and pre-aggregation so that the AI-heavy user session remains within a controlled token budget suitable for SME-focused usage.

8.2 Technical Constraints

- The preliminary MVP supports up to 3 active kiosk locations in the operational view and up to 5 candidate locations in comparison mode.

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- CSV uploads must follow a predefined schema for item, quantity, date, outlet, and sales fields.
- Forecast horizon is limited to the next 7 days in the preliminary round.
- Some integrations may use representative or partially simulated data flows when live POS APIs are not fully ready during the sprint window.

8.3 Performance Constraints

- The product is optimised for decision support, not high-frequency real-time telemetry. Dashboard freshness depends on sync cycles rather than every-second updates.
- Target response time is under 2 seconds for preloaded dashboard views and under 8 seconds for normal AI-assisted queries in demo conditions.
- Complex location comparisons may take longer because the system coordinates multiple analysis steps before returning a consolidated recommendation.

8.4 User Input & Operational Constraints

- Meaningful forecasts and recommendations depend on having sufficiently clean historical POS and inventory data.
- Users are expected to review AI outputs before acting on high-impact business decisions such as major stock purchases or new outlet commitments.
- The system is designed to support human decision-making and does not replace financial due diligence or on-site operational verification.

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9.0 Risks & Questions Throughout Development

Risk Area	Open Question	Design Intention / Mitigation
Data quality risk	What happens when POS data is missing, inconsistent, or delayed across locations?	The team must validate required fields, show sync health, and clearly mark any insight generated from incomplete data.
Forecast reliability risk	How accurate are item-level forecasts during holidays, promotions, or new-product launches?	The team must expose confidence scores, show seasonality markers, and avoid false certainty in low-history cases.
Hallucination risk	How does the system avoid unsupported explanations from GLM?	The assistant should rely on retrieved evidence, use structured outputs, and fall back to insufficient-data states when confidence is low.
Usability risk	Will the 3D shelf view stay understandable when the number of items and racks increases?	The interface should use clear colour semantics, simple filtering, and drill-down interactions rather than cluttering the main view.
Expansion bias risk	How do we avoid over-trusting location recommendations built from incomplete market signals?	The location module should separate evidence, assumptions, and recommendation strength

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		so users can judge the result critically.

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